Stage 16N Inhomogeneous Plate-with-Hole Cycle-Jump Benchmark
Living Technical Report

Master Thesis Preparation

Last updated: 2026-06-06

Abstract

This living report records the geometry, numerical setup, HPC configuration, reference results, reinjection tests, diagnostic failures, and current conclusions for Stage 16N. The stage uses an inhomogeneous Abaqus plate-with-hole benchmark with a NEML-equivalent Chaboche UMAT. The full 1000-cycle non-jump reference has been completed and is used as the validation baseline for later fixed and adaptive cycle-jump simulations. The present state of the work is that the global response can be reproduced well after manual `SDVINI/SIGINI` state injection, but exact local state reproduction near the hole is not yet clean. Therefore fixed cycle jumps remain on hold.

Contents

1 Purpose and Current Status

The purpose of Stage 16N is to move from single-element or homogeneous cycle-jump demonstrations to an inhomogeneous finite-element benchmark. The model is deliberately more demanding: a plate with a central hole produces localized cyclic plasticity near the hole, so the method must be judged not only from global reaction-force loops but also from local stress and state-variable evolution.

Table 1: Current Stage 16N infrastructure and validation status.

Item	Status
1000-cycle full non-jump reference	Completed
16-CPU production policy	Locked
Abaqus threaded launcher	Verified
Adaptive ΔN table	Completed
Exact reinjection scaffold	Prepared
HPC mail policy	Fixed
B0-1 exact reinjection solver run	Completed
B0-1 scientific result	Review / conditional pass
B0 initialization-only audit	Completed
B0-2 / B0-3 exact reinjection jobs	On hold
Fixed cycle-jump validation	On hold
Adaptive cycle-jump workflow	On hold

2 Geometry and Model Definition

The Stage 16N model is a three-dimensional plate-with-hole benchmark. It is intended to create localized cyclic evolution near the hole while retaining a global hysteresis response that can be compared across full-reference, exact-reinjection, fixed-jump, and adaptive-jump runs.

Table 2: Stage 16N model geometry and mesh summary.

Quantity	Value
Plate length	20.0
Plate height	10.0
Plate thickness	1.0
Central hole radius	1.0
Mesh spacing	$\Delta x = \Delta y = 0.25$
Element type	C3D8
Nodes	6642
Elements	3148
Hole-ring elements	60
Material model	NEML-equivalent Chaboche UMAT
State variables	27 SDVs

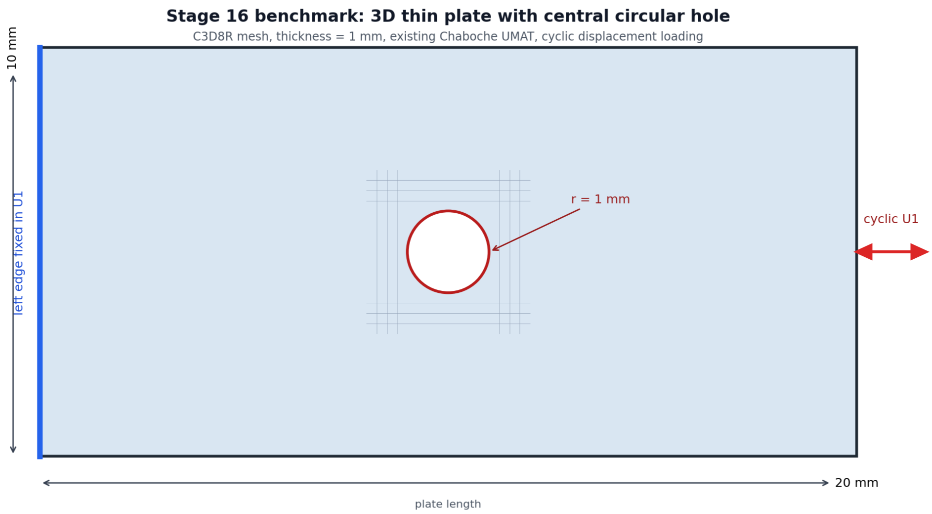

Figure 1: Plate-with-hole geometry used for the Stage 16N benchmark.

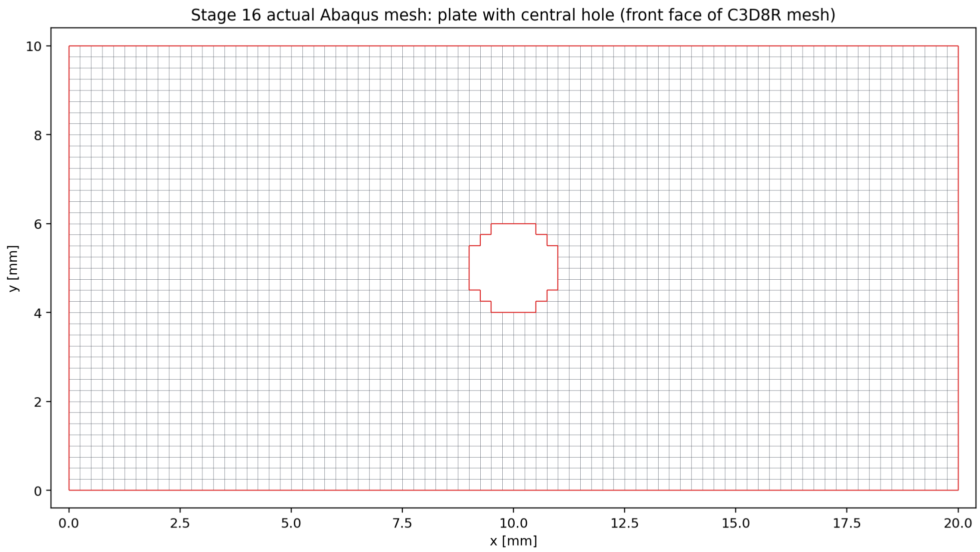

Figure 2: Actual structured C3D8 mesh with the element-wise approximation of the central hole.

2.1 Boundary Conditions and Loading

The left edge is fixed in the axial direction. Two anchor nodes constrain rigid-body modes. The right edge receives a cyclic axial displacement amplitude of ± 0.10. The full reference contains 1000 full non-jump cycles. Selected field output is stored at cycles 1, 2, 10, 50, 100, 250, 500, 750, and 1000.

3 HPC and Production Policy

The first corrected 1000-cycle parallel reference used 30 OpenMP threads, but PBS accounting showed that the average effective CPU usage was far below the requested 30 cores. A 16-CPU verification run was therefore used to lock a more resource-efficient production policy.

Table 3: Locked Stage 16N production setting.

Setting	Value
Production default	1 MPI rank × 16 OpenMP threads
PBS request	`select=1:ncpus=16:mpiprocs=1:ompthreads=16`
Abaqus launch	`cpus=16 mp_mode=threads`
Expected message-file line	`1 MPI RANK x 16 THREADS`

This policy is not claimed to perfectly saturate all 16 CPUs. It is a resource-efficient compromise that prevents the earlier serial fallback and avoids reserving 30 CPUs for a workload that did not use them efficiently.

3.1 Mail Notification Policy

Future PBS jobs must explicitly request mail on abort, begin, and end:

```
#PBS -m abe
#PBS -M pr21vyci@mailserver.tu-freiberg.de
```

This was added because PBS accounting previously showed `Mail_Points = a`, which only requests abort/failure mail.

4 Full 1000-Cycle Non-Jump Reference

The full non-jump reference was completed using Abaqus/Standard with the corrected threaded launcher.

Table 4: Completed 1000-cycle reference run.

Quantity	Value
PBS job	1335408.mmaster02
Abaqus job	`stage16n_parallel_max_reference_1000cycles`
Walltime	17:56:43
CPU time	178:03:42
Requested resources	30 CPUs, 135 GB
Abaqus parallelism	1 MPI rank × 30 threads
Completed cycles	1000
Last partial cycle	none

Table 5: Reference evolution from cycle 1 to cycle 1000.

Quantity	Cycle 1	Cycle 1000	Change
`RF1_max`	2356.834	2908.244	+23.40%
`RF1_min` magnitude	2509.712	3074.823	+22.52%
`loop_area_abs`	433.822	577.068	+33.02%
`HOLE_RING_MISES_MAX`	398.503	492.827	+23.67%
`HOLE_RING_SDV1_MAX`	0.0561	26.0519	–
`HOLE_RING_SDV8_MAX`	0.2996	92.4455	–
`HOLE_RING_SDV11_MAX`	7.9019	61.3076	–

The evolution clearly exceeds the original continuation thresholds for global loop quantities and local hole-ring state variables. This confirms that the plate-with-hole benchmark is meaningful for cycle-jump validation.

5 Adaptive ΔN Table

The adaptive ΔN table was computed from the completed 1000-cycle reference. It uses dense global cycle metrics and selected local field-output anchors. The controlling variable is the most sensitive monitored quantity, not only the global reaction force.

Table 6: Conservative adaptive ΔN estimates from the 1000-cycle reference.

Base cycle	Next anchor	Target	ΔN	Controlling variable
1	2	2	1	`HOLE_RING_SDV8_MAX`
2	10	3	1	`HOLE_RING_SDV8_MAX`
10	50	14	4	`HOLE_RING_SDV8_MAX`
50	100	61	11	`HOLE_RING_SDV1_MAX`
100	250	124	24	`HOLE_RING_SDV1_MAX`
250	500	299	49	`HOLE_RING_SDV1_MAX`
500	750	575	75	`HOLE_RING_SDV1_MAX`
750	1000	849	99	`HOLE_RING_MISES_MAX`

The table shows that local state variables near the hole strongly restrict the allowable jump size. This supports the decision that any later fixed/adaptive jump validation must include local metrics, not only global RF loops.

6 Stage 16N-B: Exact Reinjection Verification

Before any real cycle jump is allowed, the workflow must prove that stress and state-variable fields can be transferred into a new Abaqus run. The intended exact reinjection cases are:

- B0-1: exact cycle 100 state, continue normally to cycle 250.
- B0-2: exact cycle 250 state, continue normally to cycle 500.
- B0-3: exact cycle 500 state, continue normally to cycle 1000.

B0-2 and B0-3 remain on hold until the B0-1 local mismatch is understood.

6.1 State Reader Implementation

The initial CSV preload design failed under threaded Abaqus because multiple threads attempted to load the same file concurrently. The state-transfer implementation was therefore changed to a direct-access binary reader. Each record contains:

```
S1, S2, S3, S4, S5, S6, SDV1, ..., SDV27
```

The record number is:

```
record = (NOEL - 1) * 8 + NPT
```

On the Abaqus/Intel Fortran environment used here, direct-access `RECL` is interpreted in 4-byte words. Therefore the correct value for 33 double-precision values is:

```
RECL = 66
```

7 B0-1 Exact Cycle-100 to Cycle-250 Reinjection

B0-1 completed the Abaqus solver successfully. PBS reported exit status 2 only because the original postprocessing path failed after solver completion; extraction was rerun manually afterward.

Table 7: B0-1 run accounting and solver status.

Quantity	Value
PBS job	1336497.mmaster02
Route	exact cycle 100 state → cycle 250
PBS walltime	01:41:36
PBS CPU time	10:52:42
Requested CPUs	16
Abaqus solver	completed
Abaqus parallelism	1 MPI rank × 16 threads
Scientific status	Review / conditional pass

Table 8: B0-1 comparison against the 1000-cycle reference at cycle 250.

Quantity	Relative error
`RF1_max`	0.259048%
`RF1_min`	0.687538%
`loop_area_abs`	0.967300%
`HOLE_RING_MISES_MAX`	7.63648%
`HOLE_RING_S11_MAX_ABS`	9.58753%
`HOLE_RING_SDV1_MAX`	0.360172%
`HOLE_RING_SDV8_MAX`	10.9164%
`HOLE_RING_SDV11_MAX`	7.27213%

The global response is excellent, but the local hole-ring stress and state-variable errors are too large for a clean exact-restart claim.

8 B0-1 Local Diagnostic Review

A pointwise hole-ring diagnostic was run to determine whether the high local error was only a maximum-location artifact. It was not. The reference and reinjection maxima for `SDV8` occurred at the same element and integration point.

Table 9: B0-1 SDV8 argmax diagnostic at cycle 250.

Quantity	Value
Reference SDV8 argmax	element 1242, IP 7
Reinjection SDV8 argmax	element 1242, IP 7
Same argmax location	true
Reference value	91.1466598511
Reinjection value	81.1967315674
Argmax relative error	10.9164%

This made B0-1 a real local mismatch rather than a harmless maximum-location shift.

9 B0 Cycle-100 Initialization-Only Audit

The initialization audit was created to answer whether the local mismatch already exists immediately after exact cycle-100 initialization, before any cycle-100 to cycle-250 continuation.

Table 10: Cycle-100 binary reader and initialization audit.

Quantity	Value
CSV-vs-binary maximum absolute error	3.55×10^{-15}
Common hole-ring element/IP records	480
`SDV8` median relative error	12.7022%
`SDV8` 95th percentile relative error	150.665%
`SDV8` maximum relative error	479.090%
`SDV8` argmax location	same element/IP, 1242/7

The binary extraction and reader pipeline is correct to machine precision. Therefore the local mismatch is not caused by the Python binary writer or direct-access file format. The mismatch appears during manual Abaqus initialization and the tiny equilibrium step.

9.1 Initialization Audit Interpretation

The most likely interpretation is that `SDVINI/SIGINI` inject stress and state variables correctly, but the manual state-initialized model does not reconstruct the compatible displacement, strain, solver-history, and full equilibrium state of the original reference analysis. Strain-like local state variables such as `SDV8` are therefore adjusted during the first equilibrium solve. This means B0-1 is useful as a practical state-initialized continuation test, but it is not a mathematically exact Abaqus restart for local fields.

10 Failures and Fixes

Table 11: Important failures and fixes recorded in Stage 16N.

Issue	Observation	Fix / conclusion
Accidental serial-like run	A 22-hour run reached only 592 full cycles.	Parallel launcher corrected; later 1000-cycle reference completed.
30-CPU over-allocation	30-thread run averaged about 10 effective cores.	16 CPUs selected as production compromise.
Missing start/end email	PBS accounting showed `Mail_Points = a`.	Added `#PBS -m abe` and mail user to Stage 16N submit scripts.
CSV state-reader race	Threaded Abaqus calls attempted concurrent CSV preload.	Replaced with direct-access binary state reader.
Wrong direct-access record length	Abaqus/Intel runtime expected word-based `RECL`.	Set `RECL = 66` for 33 double values.

Issue	Observation	Fix / conclusion
Postprocessing path failure	B0-1 solver completed but PBS exit status was 2.	Fixed extractor path and reran extraction manually.
B0-1 local mismatch	Global errors under 1%, local `SDV8` max error 10.9164%.	Added pointwise diagnostic and initialization-only audit.
Initialization local mismatch	Binary audit passed, but `SDV8` mismatch appeared immediately.	Concluded manual `SDVINI/SIGINI` is not a mathematically exact local restart.

11 Current Conclusion

Stage 16N has successfully produced a full 1000-cycle non-jump reference and a conservative adaptive ΔN table. The inhomogeneous plate-with-hole benchmark is scientifically useful because both global hysteresis quantities and local hole-ring state variables evolve significantly. The HPC workflow is now resource-controlled with a 16-CPU production policy and explicit start/end/failure mail notifications.

The important unresolved issue is exact local reinjection. B0-1 proves that the manual `SDVINI/SIGINI` workflow can run and reproduce the global force-displacement response well, but it does not reproduce local strain-like state variables exactly. The initialization-only audit shows that this local mismatch appears immediately during manual Abaqus initialization/equilibration, even though the binary reader is exact.

12 Next Work

The next decision must be methodological, not simply computational. B0-2, B0-3, fixed jumps, and adaptive jumps remain on hold until the reinjection limitation is handled. The recommended next options are:

1. Add a true Abaqus restart comparison from cycle 100 to cycle 250, if restart data are available.
2. Investigate whether a displacement/strain-field initialization or predefined field strategy can reduce the local `SDV8` mismatch.
3. If manual `SDVINI/SIGINI` is retained as the practical state-jump method, document that it is not a mathematically exact restart and use local equilibration error as part of the cycle-jump error budget.
4. Only after this decision, run B0-2 and B0-3 or begin conservative fixed jump tests.

A Key Repository Evidence Files

- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_parallel_ma`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_adaptive_de`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_HPC_MAIL_PO`